# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – SCENARIO BASED

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|---|---|---|---|
| **Course Code:** | DSA0307 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO2 – Manipulation | **Assessment:** | Assessment Tool 2 – SCENARIO BASED |
| **Weightage:** | 30% of CIA | | |
| **CO5 Statement:** | Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3) | | |
| **Student Name:** | C Nithisha | **Reg. No.:** | 192424375 |
| **Section / Batch:** | 2024 | **Date:** | 03-08-2026 |

### Code of Conduct

*I _C Nithisha 192424375 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _C Nithisha_____

### Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

| Section / Topic | Focus | Q Nos |
|---|---|---|
| English Word Classes, Tagsets for English, Part of Speech Tagging | Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing. | Q1 |
| English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging | Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation. | Q2 |
| Counting Words, English Word Classes, Part of Speech Tagging | Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization. | Q3 |
| Rule-Based Part of Speech Tagging, Transformation-Based Tagging | Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts. | Q4 |
| Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging | Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations. | Q5 |

## Annexure A – SCENARIO BASED

**1. A large-scale search engine indexing system processes user queries containing word variants such as “analysing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.**

**Answer:**

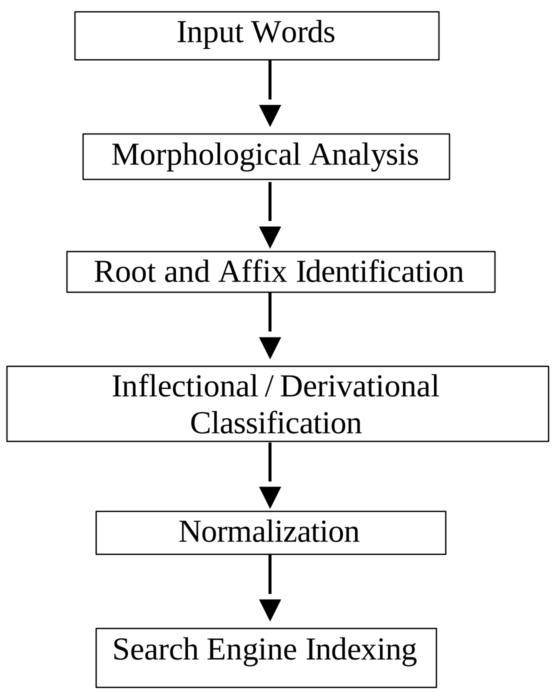
**Scenario**

A large-scale search engine indexing system processes user queries containing word variants such as "analysing", "analysis", and "analytical". The system must improve retrieval accuracy by grouping semantically related terms under a unified representation.

**Morphological Processing Approach**

The morphological processing approach analyses each word by separating it into its root and affixes. It then identifies whether the word is inflectional or derivational and normalizes all related words to a common base form. This enables the search engine to retrieve relevant documents even when different word forms are used.

**Process**



**Morphological Breakdown**

| Input Word | Root | Affix | Type |
|---|---|---|---|
| analysing | analyse | ing | Inflectional |
| analysis | analyse | sis | Derivational |
| analytical | analyse | tical | Derivational |

**Normalized Representation**

| Input Word | Normalized Form |
|---|---|
| analysing | analyse |
| analysis | analyse |
| analytical | analyse |

**Conclusion**

The words are successfully decomposed into their root and affixes. Analysing is an inflectional form, while analysis and analytical are derivational forms. All variants are normalized to analyse, improving search engine indexing and retrieval.

**2. An AI-based sentiment analysis platform receives product reviews containing words like "disagree", "agreement", and "agreeable". The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.**

**Scenario**

An AI-based sentiment analysis platform receives product reviews containing "disagree", "agreement", and "agreeable".

**Morphological Parsing Strategy**

The parser identifies prefixes, suffixes, and the base form of each word. It analyzes derivational changes to understand how the meaning changes while maintaining semantic consistency during sentiment analysis.

**Process**

Input Words
↓
Morphological Parsing
↓
Prefix Identification
↓
Root Identification
↓
Suffix Identification
↓
Semantic Interpretation

**Morphological Breakdown**

| Input Word | Prefix | Root | Suffix |
|---|---|---|---|
| disagree | dis | agree | — |
| agreement | — | agree | ment |
| agreeable | — | agree | able |

**Classification**

| Word | Transformation |
|---|---|
| disagree | Derivational |
| agreement | Derivational |
| agreeable | Derivational |

**Semantic Interpretation**

- disagree – expresses the opposite meaning of *agree*.
- agreement – represents the state or act of agreeing.
- agreeable – describes a pleasant or cooperative quality.

**Normalized Representation**

All words are normalized to:

Agree

**Conclusion**

The parser correctly identifies prefixes, suffixes, and root words while preserving semantic meaning. All forms are mapped to the common base form agree.

**3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown**

**Scenario**

A text analytics pipeline processes "govern", "government", and "governance".

**Morphology-Based Normalization Mechanism**

The normalization mechanism decomposes each word into its root and suffix, identifies derivational layers, and maps all forms to a common base representation for topic modeling and clusterin

**Process**

Input Words
↓
Morphological Analysis
↓
Root and Suffix Identification
↓
Derivational Layer Detection
↓
Normalization
↓
Topic Modeling

**Morphological Breakdown**

| Input Word | Root | Suffix |
|---|---|---|
| govern | govern | — |
| government | govern | ment |
| governance | govern | ance |

**Derivational Layers**

| Word | Type |
|---|---|
| govern | Base Form |
| government | Derivational |
| governance | Derivational |

**Normalized Representation**

| Input Word | Normalized Form |
|---|---|
| govern | govern |
| government | govern |
| governance | govern |

**Conclusion**

The morphology-based normalization mechanism maps all word forms to the common root govern, improving document clustering and topic modelling.

**4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.**

**Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.**

**Analyse how prefix addition and suffix transformation affect meaning and word class.**

**Apply the process to decompose each word and map them to a unified base representation.**

**Derive the morphological structure and normalized root for each input word.**

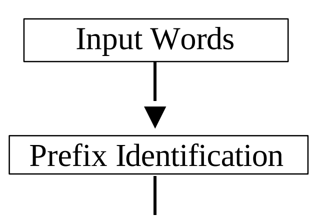
**Scenario**

A document classification system processes "activate", "activation", and "reactivation".

**Morphological Parsing Strategy**

The strategy identifies prefixes, suffixes, and root words while distinguishing derivational layers. It analyzes how affixes modify the meaning and grammatical category of each word

**Process**



↓

Root Identification

↓

Suffix Identification

↓

Derivational Analysis

↓

Normalization

**Morphological Breakdown**

| **Input Word** | **Prefix** | **Root** | **Suffix** |
|---|---|---|---|
| activate | — | activate | — |
| activation | — | activate | ion |
| reactivation | re | activate | ion |

**Meaning and Word Class**

| **Word** | **Effect** |
|---|---|
| activate | Base verb |
| activation | Noun formed from the verb |
| reactivation | Indicates activation again |

**Normalized Representation**

| **Input Word** | **Normalized Root** |
|---|---|
| activate | activate |
| activation | activate |
| reactivation | activate |

**Conclusion**

The parsing strategy successfully identifies prefixes, suffixes, and root forms. All words are normalized to activate, improving semantic grouping and indexing.

**5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.**

**Identify suffix patterns and map all variations to a common base form without altering meaning.**

**Apply the process to extract root forms and classify each transformation.**

**Determine the morphological breakdown and normalized output for each word**

**Scenario**

A search optimization module processes "create", "creates", and "creating".

**Morphology-Based Normalization Approach**

The approach focuses on inflectional transformations by identifying suffix patterns and mapping all grammatical variations to the common base form without changing the meaning.

**Process**

Input Words
↓
Morphological Analysis
↓
Suffix Identification
↓
Inflectional Classification
↓
Normalization
↓
Search Optimization

**Morphological Breakdown**

| Input Word | Root | Suffix |
|---|---|---|
| create | create | — |
| creates | create | s |
| creating | create | ing |

**Classification**

| Word | Transformation |
|---|---|
| create | Base Form |
| creates | Inflectional |
| creating | Inflectional |

**Normalized Representation**

| Input Word | Normalized Form |
|---|---|
| create | create |

| Input Word | Normalized Form |
| --- | --- |
| creates | create |
| creating | create |

**Conclusion**

The normalization approach successfully identifies suffix patterns, classifies the grammatical transformations as inflectional, and maps all word forms to the common base form create. This improves search optimization by ensuring consistent retrieval across different grammatical variations.

## Rubrics for Calculation

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |

| Q. No. / Criteria Level | Understanding of Scenario | Identification of Key Issues | Application of Concepts | Decision Making | Clarity of Response | Sub. Total | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |
| 5 | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | | |
| Signature: ________________ | Signature: ________________ | Signature: ________________ |
| Date: ____________________ | Date: ____________________ | Date: ____________________ |